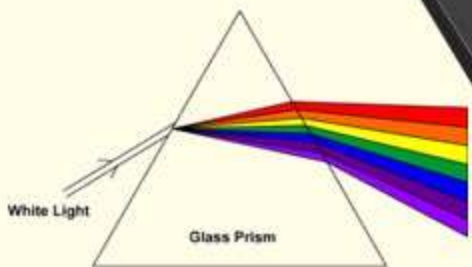


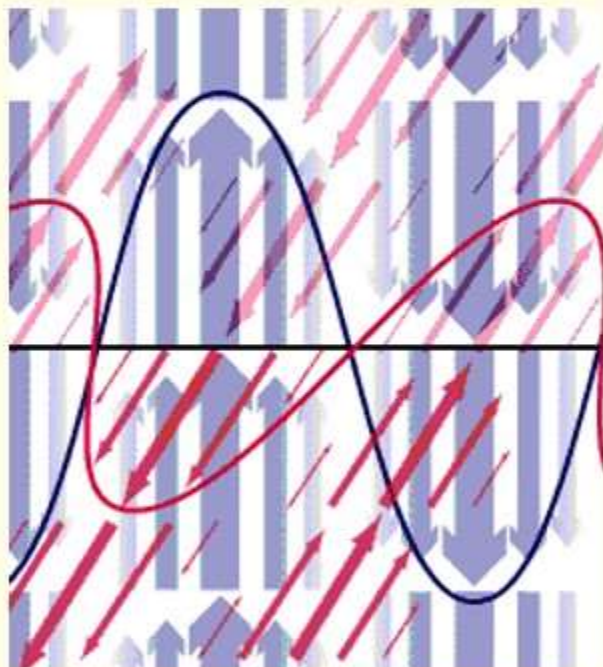
PHYSICS

OPTICS

MODULE - 5



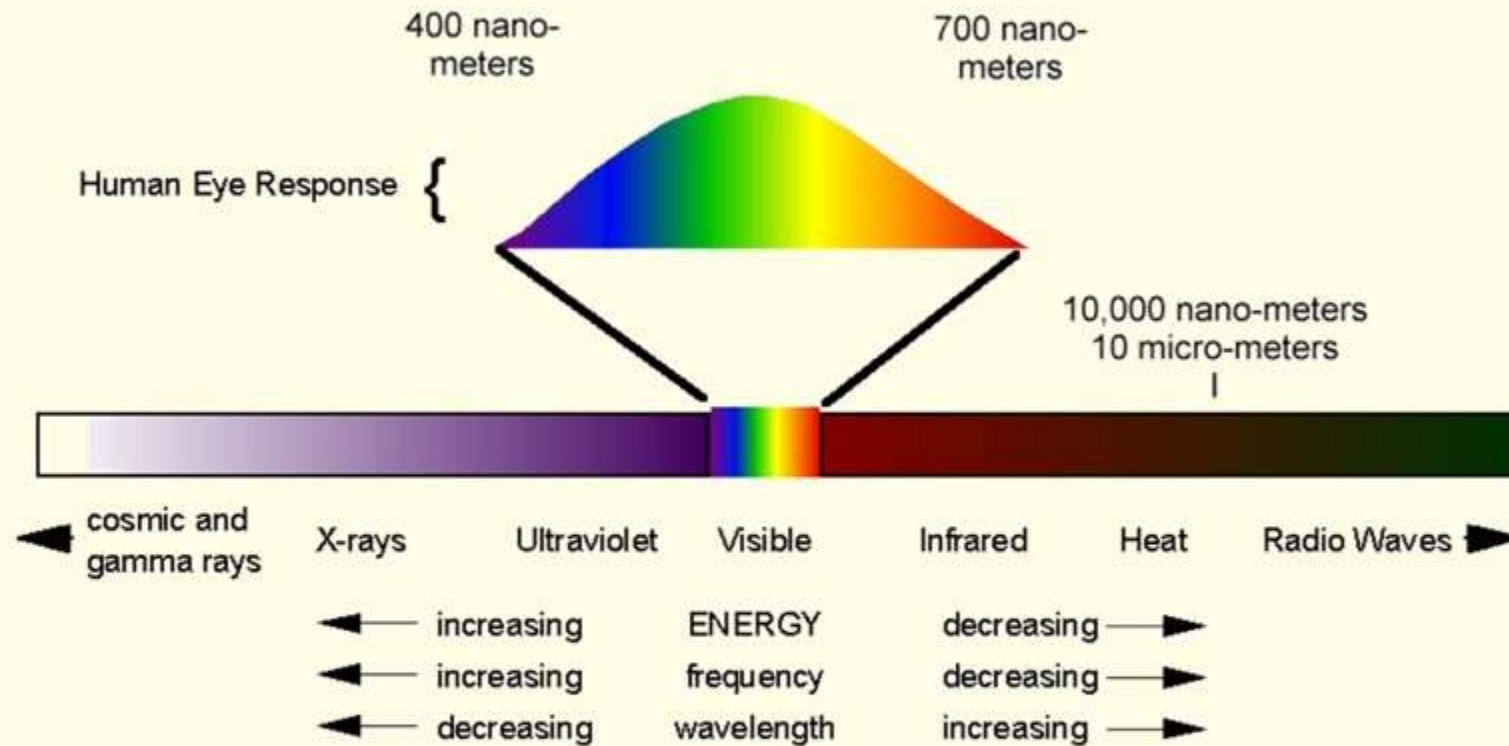
LIGHT



- Light is a form of energy that is propagated by **electromagnetic waves**.
- The magnetic and electric field is **perpendicular to each other**.
- It is a form of energy that reaches Earth enables us to see.
- The scientific study of light is called OPTICS.

Electromagnetic wave is formed when an electric field (blue arrows) couples with a magnetic field (red arrows).

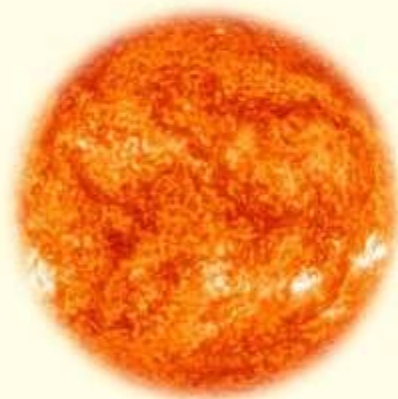
LIGHT – IN THE ELECTROMAGNETIC SPECTRUM



- In the entire range of frequencies and wavelengths of electromagnetic radiation, our eye can perceive a very small part; called as the visible spectrum.
- We can perceive **400nm to 700nm wavelength** of the spectrum with a frequency of **430 – 770 THz**.

LIGHT THAT REACHES US

- Light travels by radiation whose speed is 3×10^8 m/s.
- It takes 8 min 19 s to reach on the Earth from the Sun.
- The light reflected from the Moon takes 28 s to reach Earth.



LIGHT – THREE TYPES OF BEHAVIOR ON HITTING A SURFACE



ABSORBED

If an object absorbs all the light falling on it, then it will appear perfectly black.
E.g. a blackboard, any dark surface.



TRANSMITTED

An object is said to transmit light if it allows light to pass through it and such objects are transparent.
E.g.: a clear glass.



REFLECTED

If an object sends back light rays falling on its surface, then it is said to have reflected the light.
E.G mirror, reflecting sunglasses etc

LIGHT – HOW WE PERCEIVE IT?



A DARK ROOM



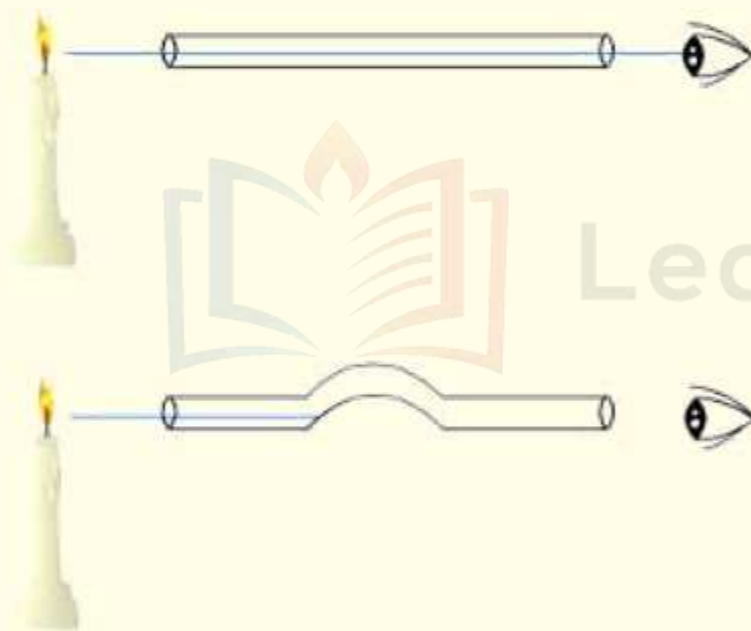
HEADLIGHT OF A CAR



A LIGHT- HOUSE

What do all the above cases suggest about light?

LINEAR PROGRESSION OF LIGHT



- All the daily observation suggests that light travels in a straight line. This is called linear progression of light.
- If the path of light is obstructed by an opaque object, light cannot pass through it by bending.



HOW CAN WE SEE EVERYTHING AROUND US?



- All the objects are visible to our eye when light falls on them and get reflected back to hit our eye.
- Our brain, then processes the image to give us an idea of what we are watching.

THE DIRECTION OF LIGHT CAN BE CHANGED!



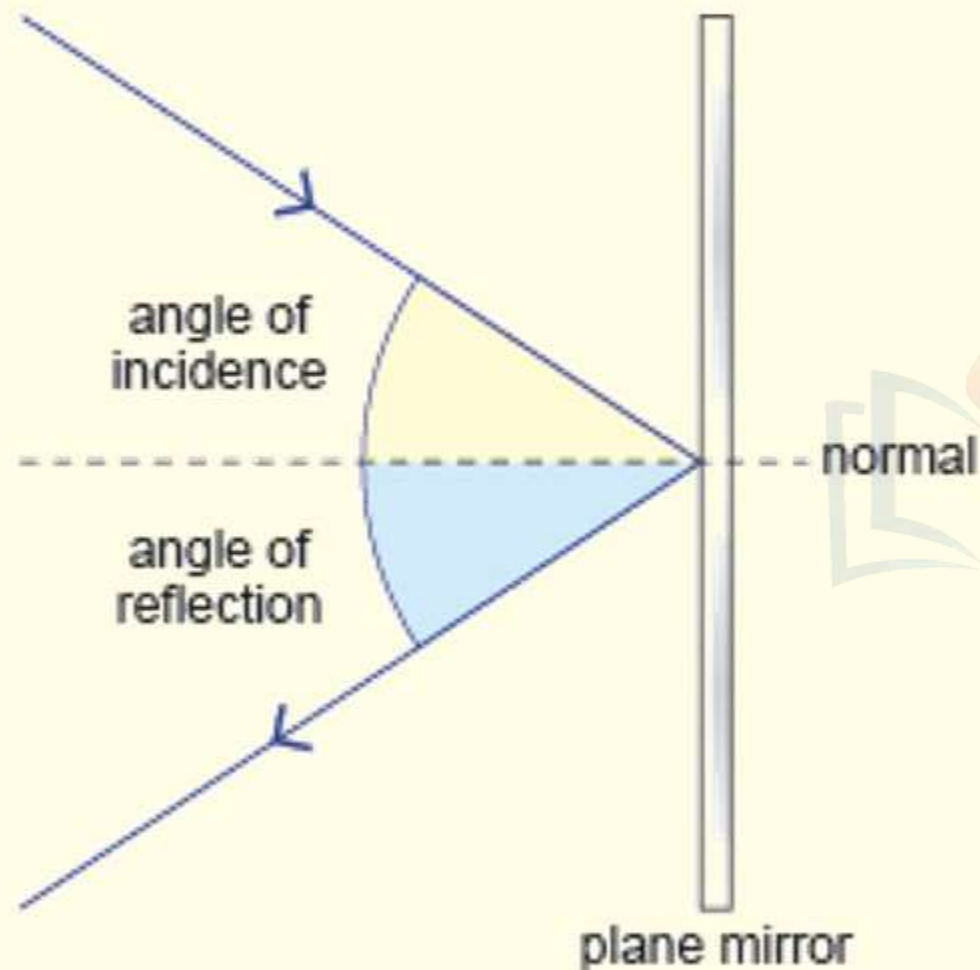
- The direction of light can be changed when it is made to shine upon a shiny surface.
- We daily observe it when we look through a shiny metal or a mirror.
- This change in the direction of light is called Reflection of light.

THE REFLECTION OF LIGHT



- Bouncing back of a **light ray** after hitting any surface is known as reflection. When a light ray is obstructed by any solid surface, it will reflect back from the surface, as it cannot pass through the surface.
A light ray is an idealized model of light, which is drawn as a straight line.

THE PHYSICS BEHIND REFLECTION OF LIGHT



INCIDENT RAY

The angle that an **incident ray** makes perpendicular to the surface at the point of incidence.

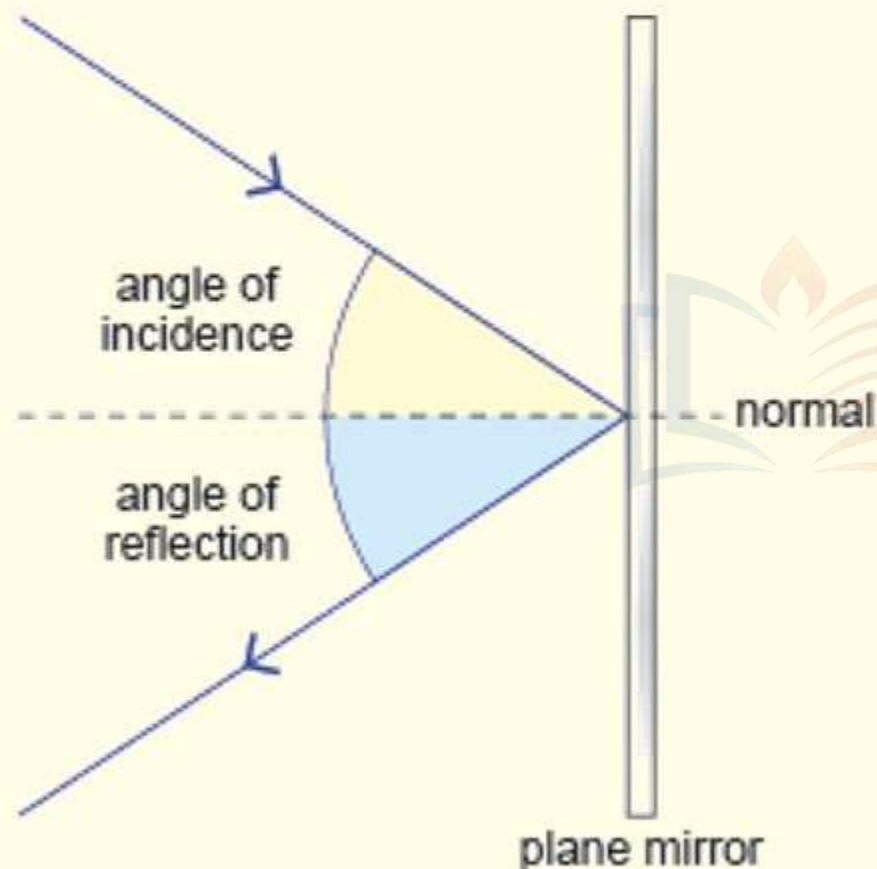
REFLECTED RAY

The angle that a ray makes perpendicular to the surface after being reflected from a surface.

NORMAL

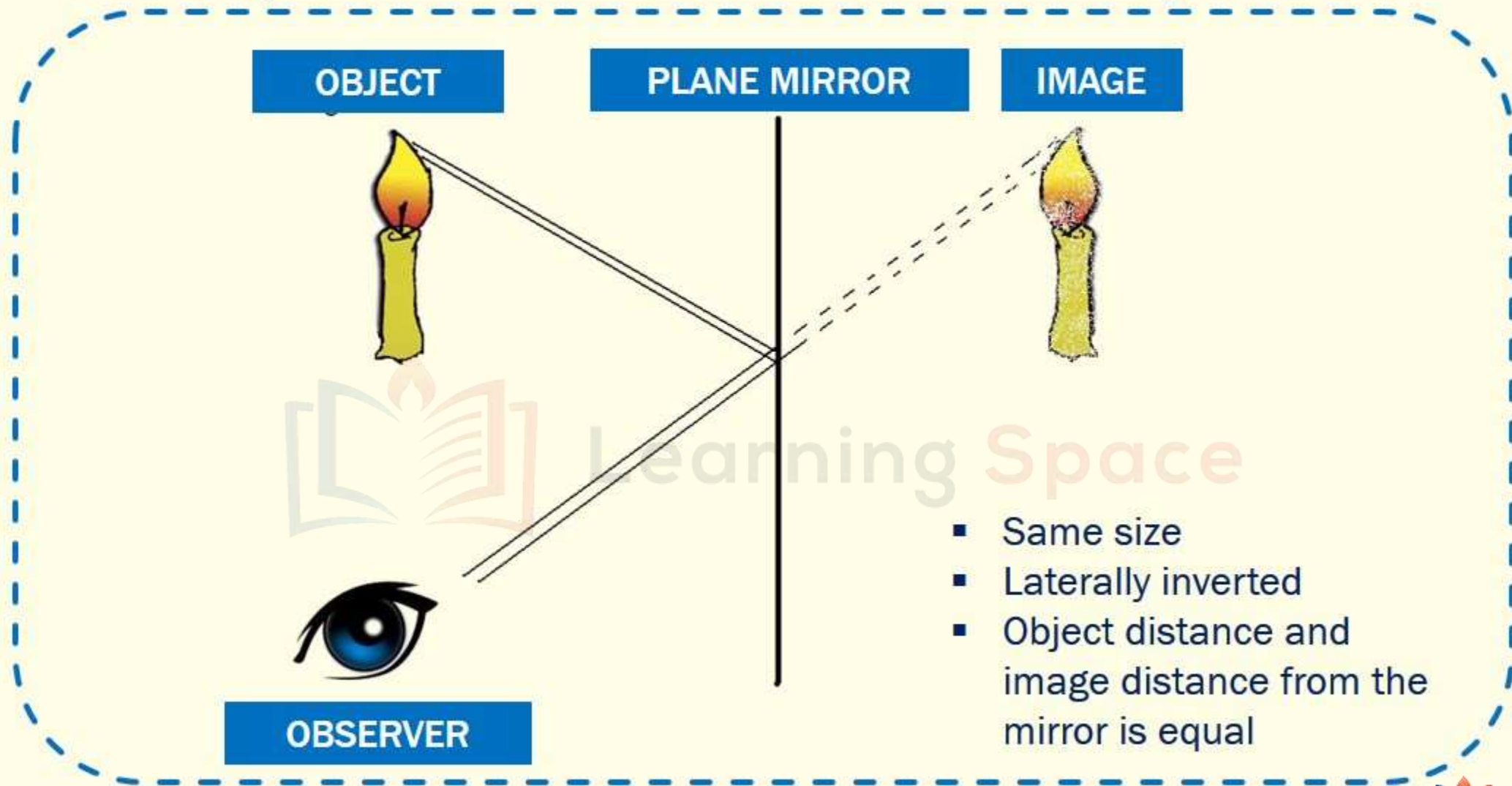
The imaginary perpendicular line drawn from the reflecting surface where the incident ray meets the reflected ray.

REFLECTION OF LIGHT – THE LAWS

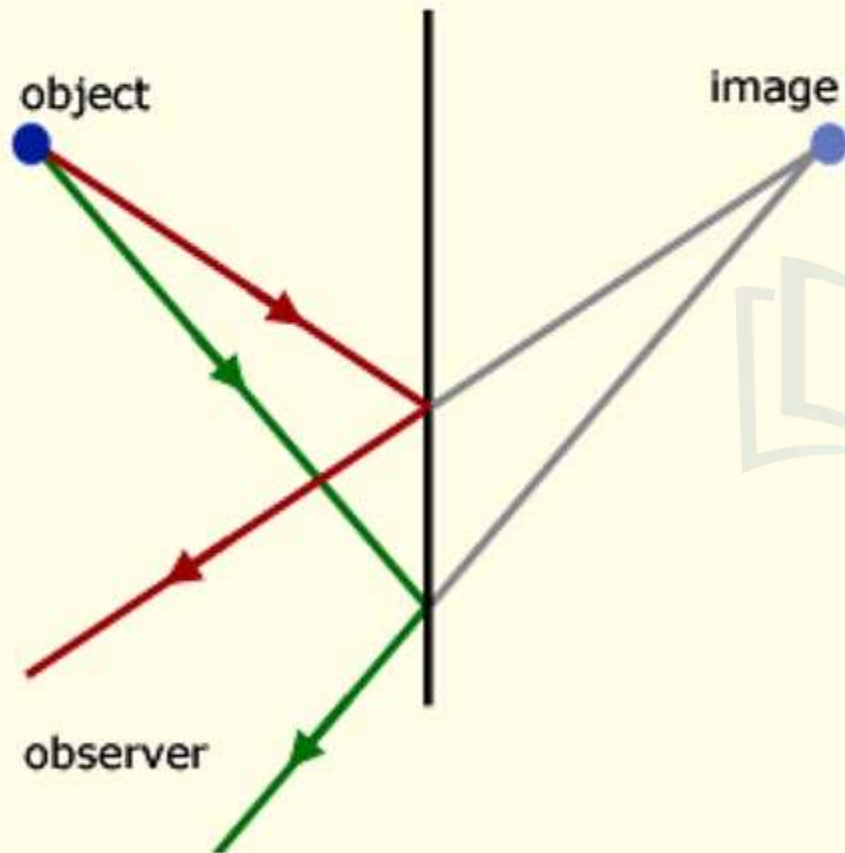


- The angle of incidence equals the angle of reflection. $\angle i = \angle r$
- The incident ray, the normal and the reflected ray all lie in the same plane.

IMAGE FORMED BY A PLANE MIRROR



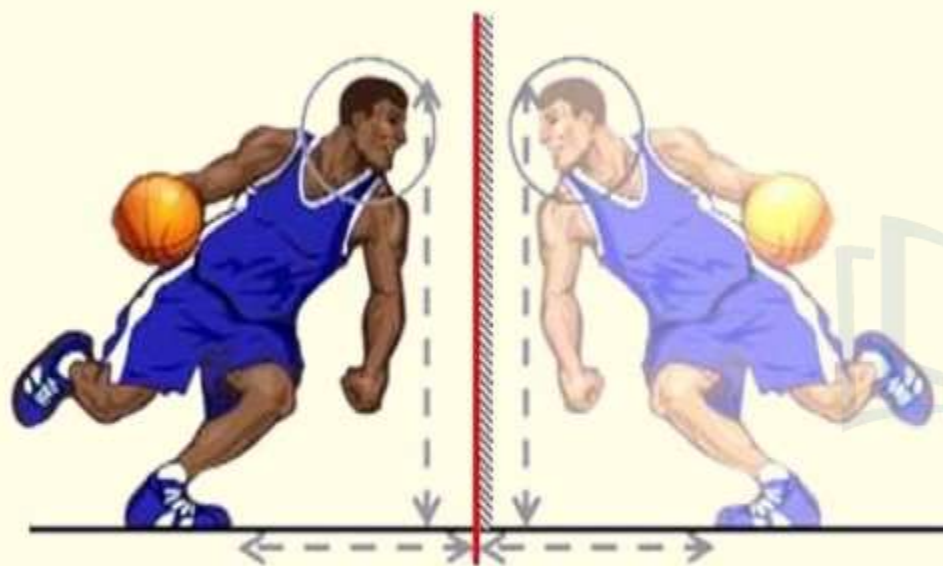
PECULARITIES OF A PLANE MIRROR IMAGE



- The distance of the object from the plane mirror and the distance of the image from the plane mirror is the same.
- If an object moves towards a plane mirror with speed v , the reflection of the object in the mirror moves towards it with a speed $2v$.
- To see his full image in a plane mirror, a person requires a mirror of at least half of his height.

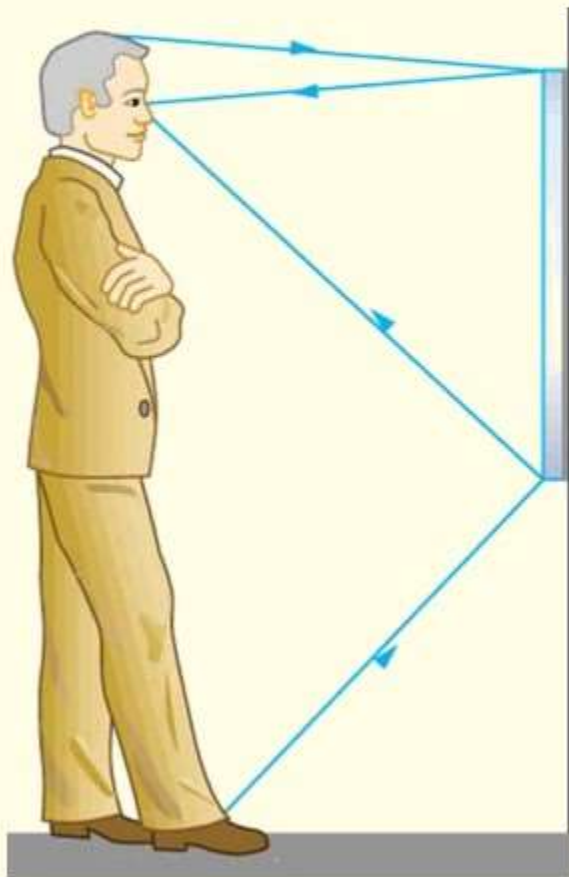


PECULARITIES OF A PLANE MIRROR IMAGE



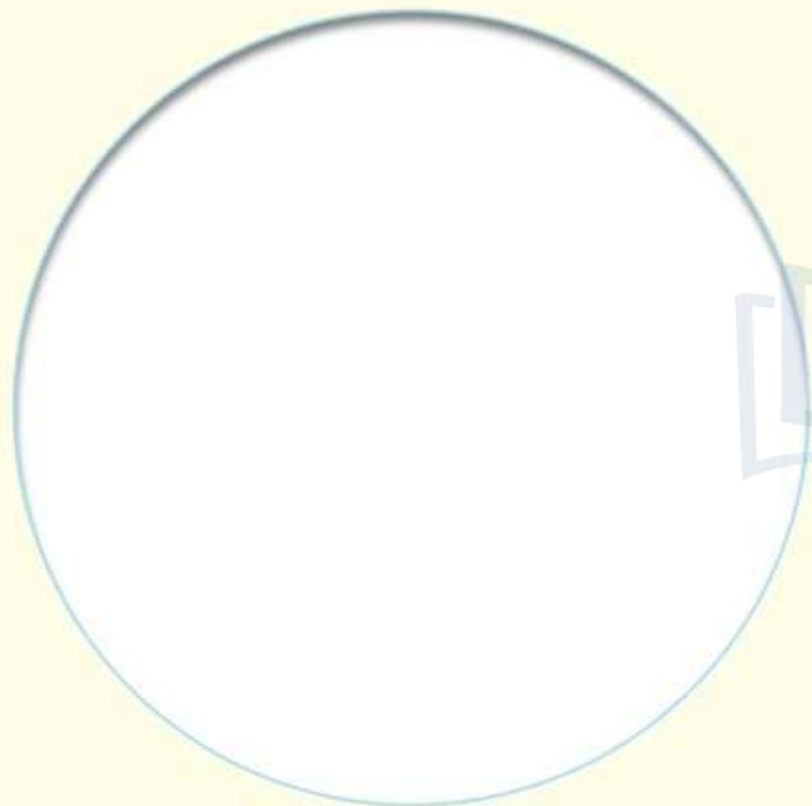
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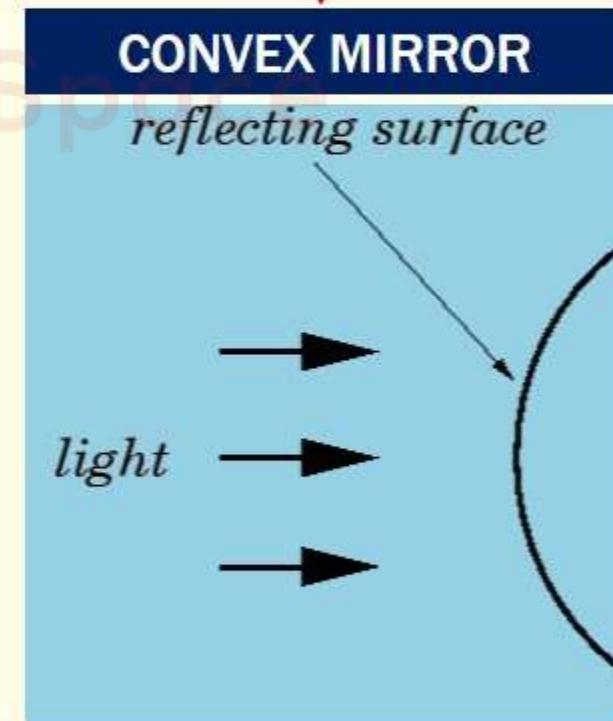
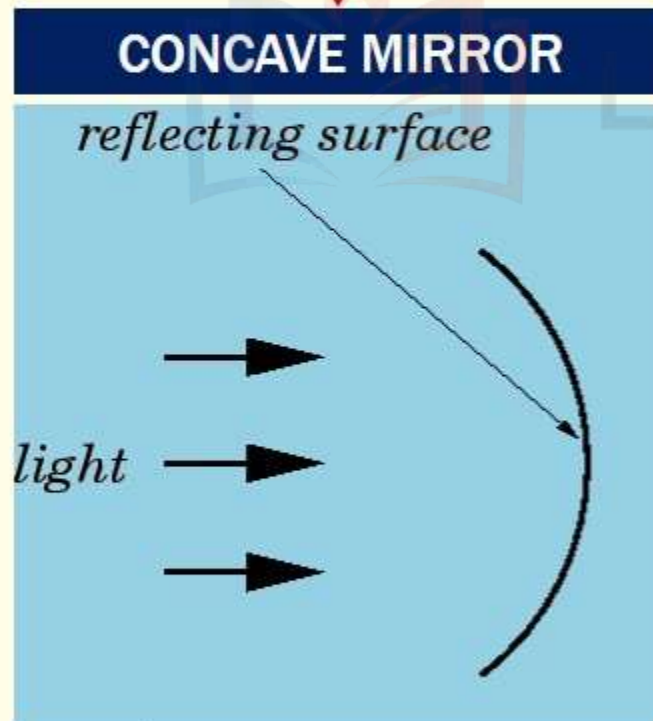
SPHERICAL MIRROR



- A spherical mirror is one which has a spherical reflecting surface.
- Spherical mirrors can have reflecting surface both on the inside as well as the outside of the sphere.
- The formation of the image of a spherical mirror is different than plane mirror.
- But, the plane mirror is also considered to be a part of a huge spherical mirror.
- If a small arc of a large spherical mirror is taken, it will behave like a plane mirror.

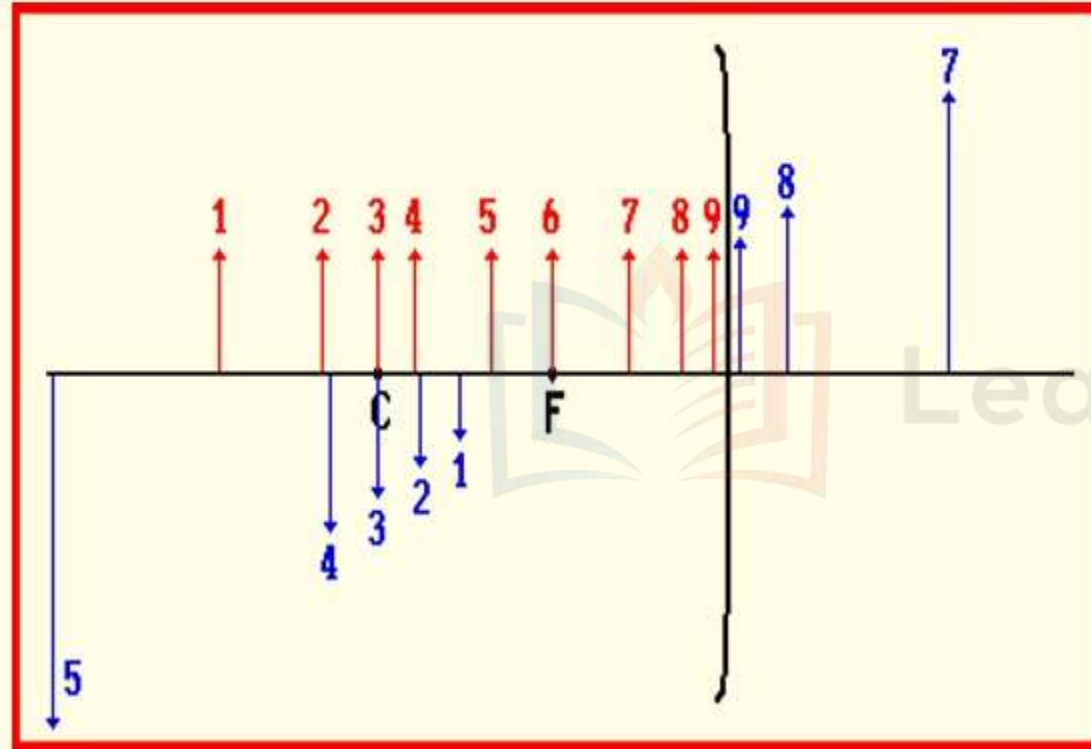


SPHERICAL MIRROR





CONCAVE MIRROR - CONVERGING



- In concave mirrors, light being reflected goes inward direction towards the focal point.
- Concave mirror usually makes a real image because it can converge light and the image can be obtained with a screen.
- From a distance, the images appear upside down, but when brought nearer image appears erect and magnified.



USES OF CONCAVE MIRRORS



HEAD LIGHT



MICROSCOPES



CAMERA FLASH



FACE MIRRORS



TORCHES



ENT MIRROR



HOW TO OVERCOME SPHERICAL ABERRATIONS IN CONCAVE MIRRORS – PARABOLIC REFLECTORS

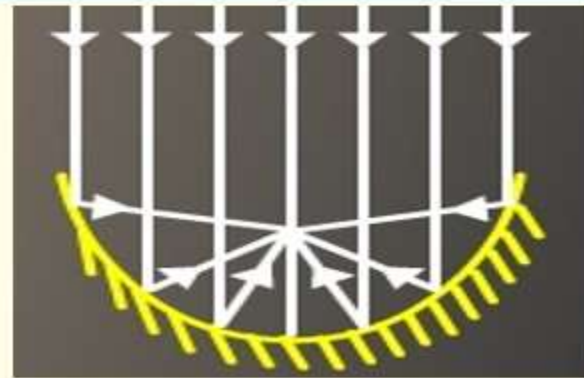
CONCAVE MIRROR

- It has a slight curvature
- In case of parallel rays that reflect from the marginal points tend to converge nearer than the focus.
- This causes spherical aberration and the image is not sharp.

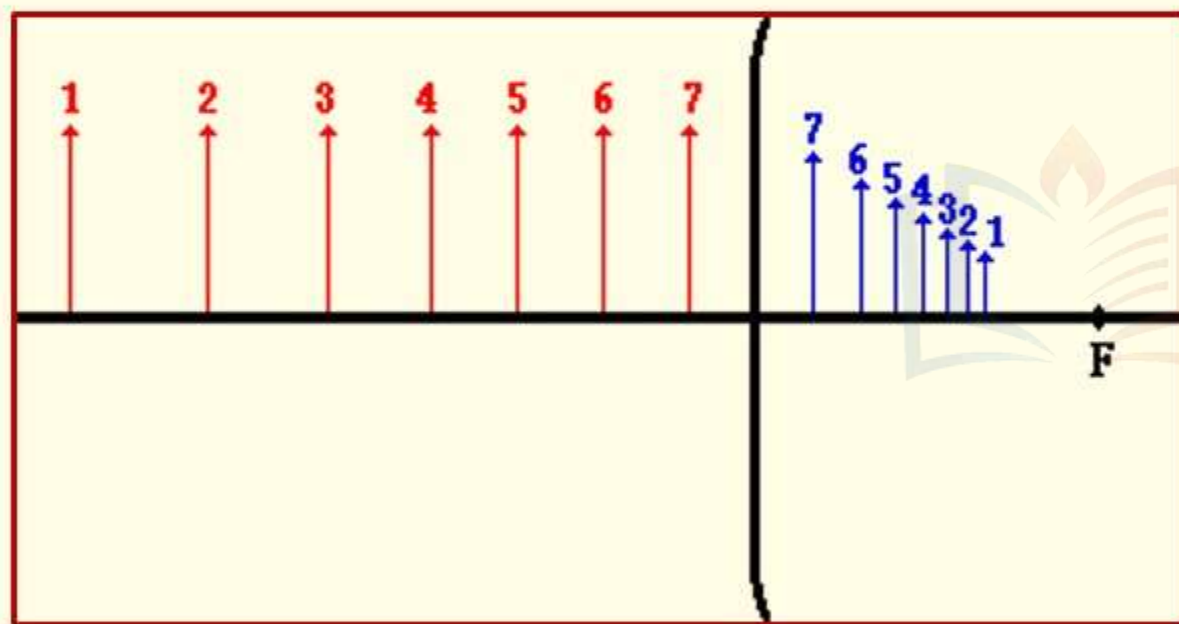


PARABOLIC MIRRORS

- It has a deeper curvature.
- In case of parallel rays the parabolic curvature converges every ray to the focus.
- This causes a sharp and crisp image at the focus.



CONVEX MIRROR



- Convex mirror diverges the light that is reflected off its surface.
- A convex mirror usually produces virtual image in which the rays seem to converge.
- They cannot be obtained in the screen.
- The image of an object appears diminished from a distance and increases its size as the object comes closer to the mirror.



CONVEX MIRROR - USES

REAR VIEW MIRROR



PARKING MIRROR



The objects in the rear view mirror are actually bigger than they appear because the convex mirror diminishes the image

WE EXPRESS OUR SINCERE THANKS FOR VIEWING THIS VIDEO

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